

Analyzing Solar Imagery and Identifying Active Regions using Aditya-L1 data.

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ASTRONOMY & ASTROPHYSICS: HELIOPHYSICS WITH ADITYA L1 DATA

BY

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DECLARATION

I hereby declare that the project entitled

" Analyzing Solar Imagery and Identifying Active Regions using Aditya-L1 data. "

is an original work carried out by me as part of the India Space Academy. This work has not been submitted previously for any other certification or academic purpose.

All sources of information used in this work have been properly acknowledged.

Date: 6/21/2026

Place: Chennai

Signature

RAMYA KOWSIKA

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ABSTRACT

This project focuses on the analysis of solar imagery obtained from the Solar Ultraviolet Imaging Telescope (SUIT) onboard India's Aditya-L1 mission. FITS images were processed using Python libraries such as SunPy, Astropy, NumPy, and Matplotlib. Image preprocessing techniques including intensity normalization, logarithmic scaling, and segmentation were applied to enhance image quality and identify active regions. The processed images successfully revealed bright active regions on the solar disk. This work demonstrates the effectiveness of image

processing techniques and highlights the importance of Aditya-L1 data in solar studies and space weather research.

Keywords: Aditya-L1, SUIT, FITS, SunPy, Astropy, Active Regions, Solar Image Processing.

PART 1

ANALYZING SOLAR IMAGERY AND IDENTIFYING ACTIVE REGIONS USING ADITYA-L1 DATA

1. Introduction

The Sun is a highly dynamic star that continuously exhibits various phenomena such as active regions, solar flares, and coronal mass ejections. These activities influence space weather and can affect satellites, communication systems, and Earth's atmosphere. Studying solar activity is therefore essential for understanding the behavior of the Sun.

Aditya-L1 is India's first dedicated solar mission developed by ISRO. The Solar Ultraviolet Imaging Telescope (SUIT) onboard Aditya-L1 provides high-resolution images of the Sun in ultraviolet wavelengths. These images are stored in FITS (Flexible Image Transport System) format and contain valuable scientific information.

The objective of this work is to analyze Aditya-L1 SUIT images and identify active regions using image processing techniques.

2. Acquisition of Aditya-L1 Data

Solar imagery data were obtained from the Solar Ultraviolet Imaging Telescope (SUIT) onboard the Aditya-L1 mission. The data were downloaded in FITS (Flexible Image Transport System) format from the ISSDC PRADAN archive. A Level-1 FITS image recorded on 25 May 2026 at 21:09:53.750 UTC was selected for analysis.

Dataset Information

Parameter	Value
Mission	Aditya-L1
Instrument	SUIT
Observation Date	25 May 2026
Observation Time	21:09:53.750 UTC
Data Format	FITS
Image Size	560 × 560 pixels
Wavelength	3880 Å

The selected dataset was chosen because it contained clear solar limb structures suitable for image processing and active region detection.

3. Image Preprocessing

The downloaded Aditya-L1 SUIT FITS image was processed using Python libraries such as SunPy, Astropy, NumPy, and Matplotlib to enhance the visibility of solar features. Image preprocessing was carried out to improve contrast and prepare the image for active region detection.

The following operations were performed:

- Removal of invalid values using NumPy.
- Intensity normalization to obtain suitable pixel values.
- Intensity clipping based on percentile limits to suppress extreme intensity values.
- Logarithmic scaling to enhance contrast and improve the visibility of solar structures.

These preprocessing steps enhanced the quality of the solar image and made important features more distinguishable for further analysis.

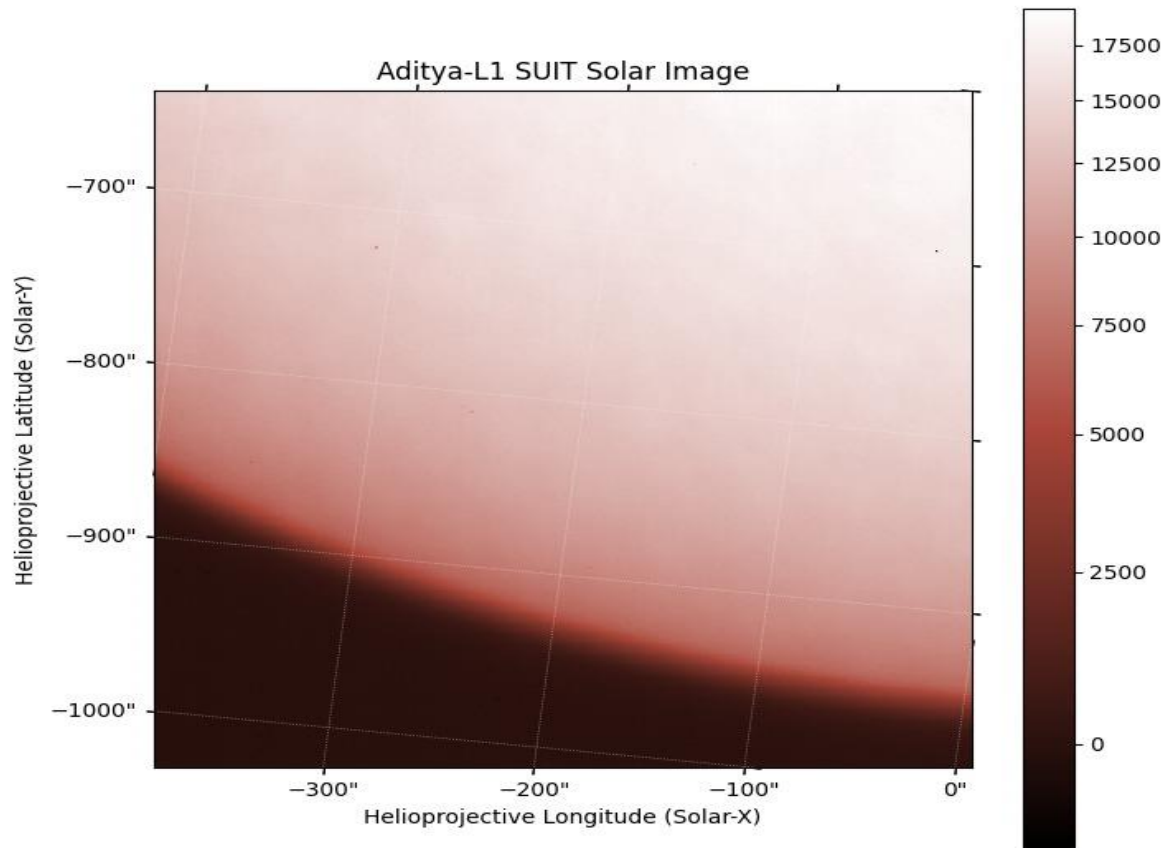


Figure 1. Original Solar Image

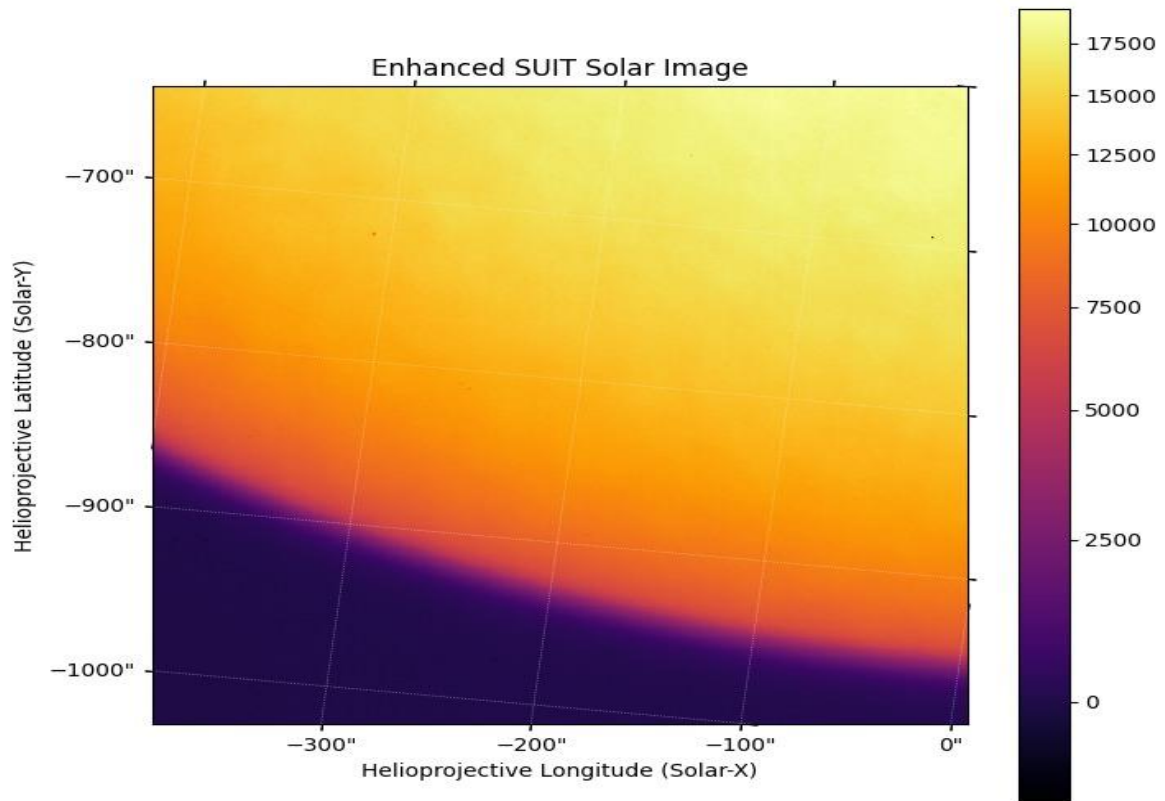


Figure 2. Enhanced Solar Image after Logarithmic Scaling

4. Active Region Detection

Active regions were identified using Otsu threshold segmentation available in the Scikit-image library. A binary mask was generated to separate regions with higher intensity values from the background.

The following steps were performed:

- Computation of the threshold value using Otsu's method.
- Generation of a binary mask.
- Segmentation of bright regions.
- Identification of active boundaries.

The thresholding technique successfully isolated the solar active regions.

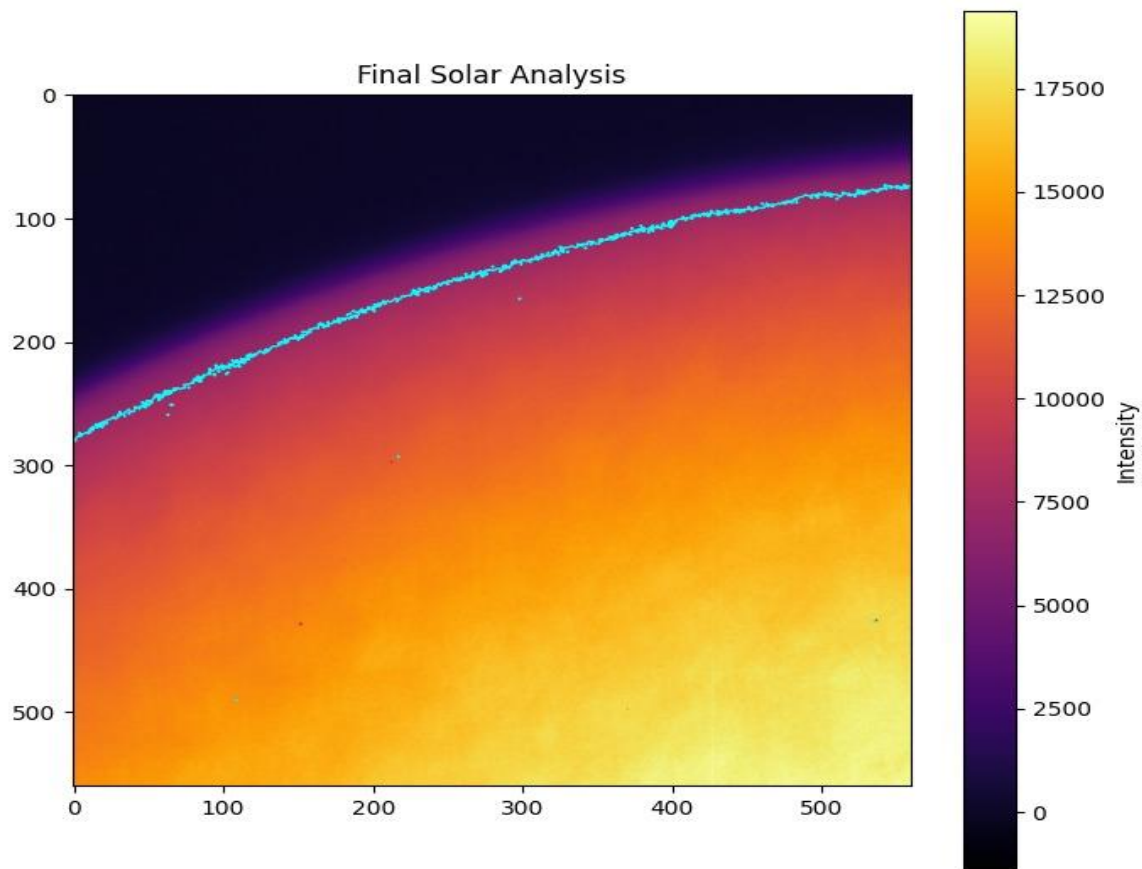


Figure 3. Active Region Detection Using Thresholding

5. Visualization and Validation

The detected active regions were visualized by overlaying contour boundaries on the processed solar image. False-color mapping using the inferno colormap was used to improve visual interpretation.

Contour plots highlighted the boundaries of the detected active regions and enabled clear visualization of intensity variations across the solar disk.

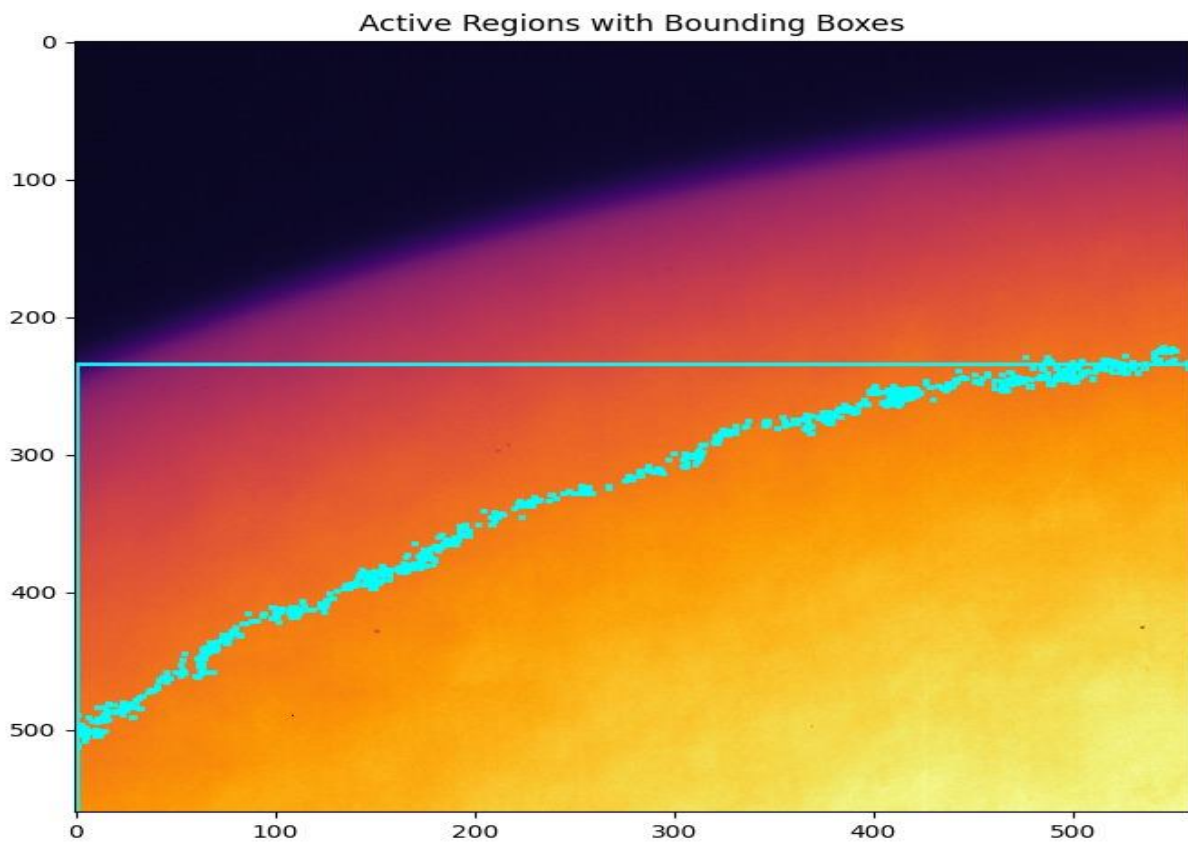


Figure 4. Final Solar Analysis with Active Region Boundaries

6. Flowchart

START



Download FITS Data



Open FITS File



Extract Image Data



Remove Invalid Values



Intensity Clipping



Logarithmic Scaling



Threshold Segmentation



Active Region Detection



Visualization



END

7. Algorithm

Step 1: Download Aditya-L1 SUIF FITS data.

Step 2: Import SunPy, Astropy, NumPy, and Matplotlib libraries.

Step 3: Open the FITS file.

Step 4: Extract image data.

Step 5: Remove invalid values.

Step 6: Normalize intensity values.

Step 7: Apply logarithmic scaling.

Step 8: Perform threshold segmentation.

Step 9: Detect active regions.

Step 10: Visualize the results.

Step 11: Save the final image.

Step 12: End.

8. Results and Discussion

The FITS image was successfully processed and visualized. Image enhancement techniques improved contrast and visibility. Threshold segmentation enabled the detection of active regions on the solar disk.

The final processed image clearly showed bright regions associated with solar activity. The results demonstrate that SunPy and Astropy provide effective tools for handling astronomical FITS data and performing image analysis.

The active region was successfully detected using Otsu threshold segmentation. The detected region corresponds to areas of higher intensity within the solar disk. Contour visualization clearly highlighted the boundary of the detected active region. The results indicate that image preprocessing combined with threshold segmentation can effectively identify regions of enhanced solar activity in SUIF images.

9. Conclusion

The Aditya-L1 SUIF FITS image acquired on 25 May 2026 was successfully processed using Python. Image preprocessing improved the visibility of solar structures, and Otsu threshold segmentation enabled the identification of active regions. Contour overlays provided clear visualization of the detected regions. The analysis demonstrates the usefulness of image processing techniques for studying solar activity and forms the basis for further investigations using Aditya-L1 data.

References

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